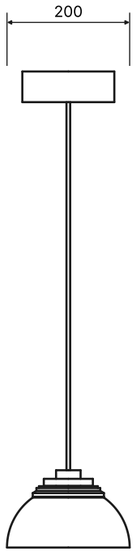
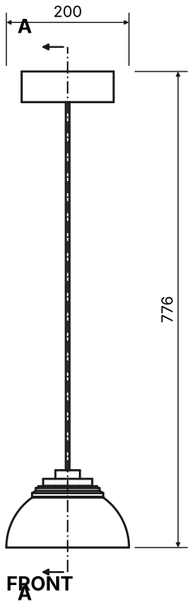
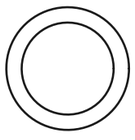
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.




FROM THE RIGHT




FRONT



FROM ABOVE

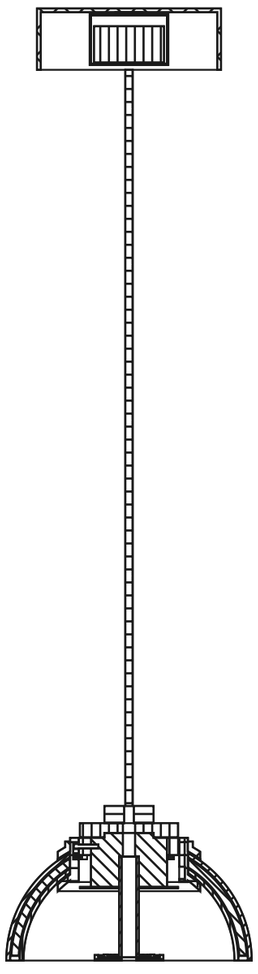
## General arrangement

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Section A-A (sheet 1): the product cut at x 0.0 and seen from the right; cut faces hatched, a different angle for each part.



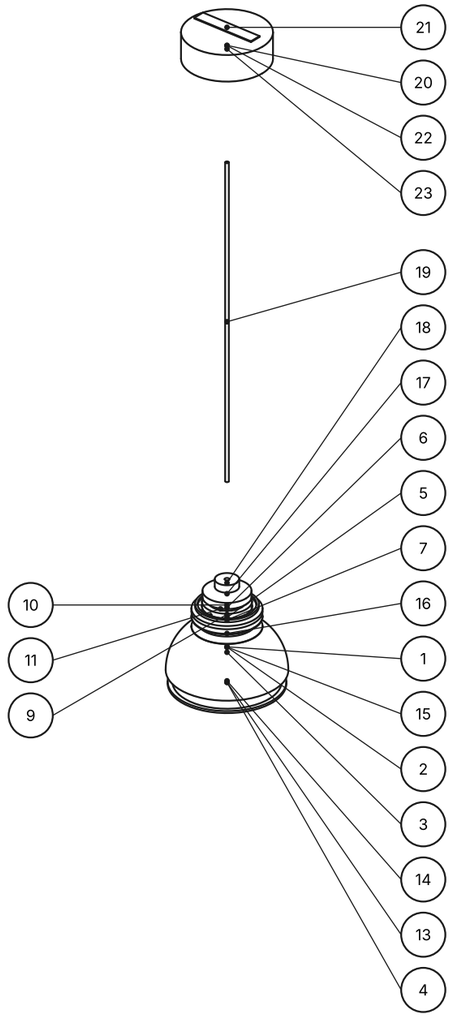
## Section A-A

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).




# Exploded view

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

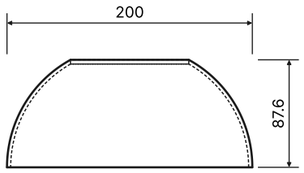
mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

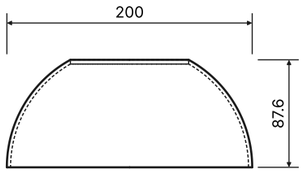
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**1. shell: formed, acrylic-tint-3-formed; 160 cm3 each; 1 for the set; 1:5**

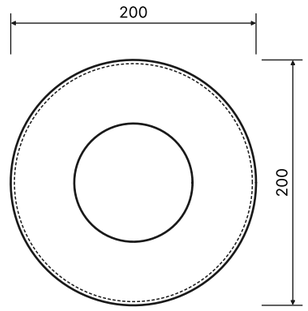
v3, the outside: 1/8 in (3 mm) stock TruCast in the line's code (Ember 2423, Poison 2092, Haze 2287, Dust 2422, Acid 9097, Ice 2069, Fog 0000, Smoke 2074), drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds; CUSTOM POURS (the owner, 2026-11-11): Dust, Acid and Ice the same dye at 3x the stock sheet's loading, Smoke at the speakers' own pour, 0.3x (about 54 % through 6 mm), not stock 2074; the one tinted layer, seen unlit, glowing lit (15 to 20 % at 3 mm for 2423 and 2092). Its mould is v2's shell mould (the ...




**FRONT**



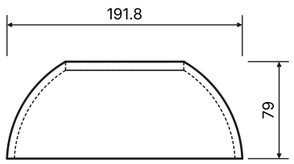

**FROM THE RIGHT**



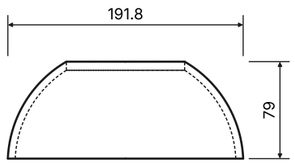

**FROM ABOVE**

**2. liner: formed, acrylic-clear-formed; 265 cm3 each; 1 for the set; 1:5**

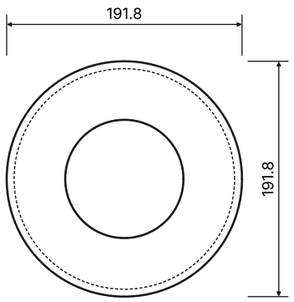
v3, the middle: 6 mm clear cast sheet (TruCast 0000), drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds: the structural shell (R 96 to 90), clamped at the crown with the others; a new mould (v2's liner mould was R 90)




**FRONT**




**FROM THE RIGHT**




**FROM ABOVE**

## Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

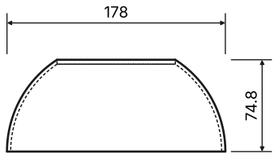
mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

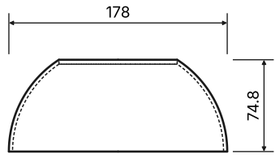
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**3. skin: formed, acrylic-opal-formed; 120 cm3 each; 1 for the set; 1:5**

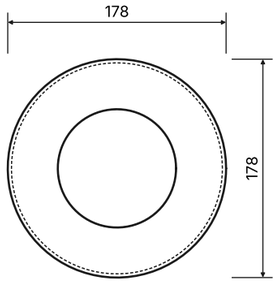
v3, the inside: 3 mm volume-diffusing (opal) cast acrylic, Perspex GS 041 or equal (T 54 %, R 39 %, ASTM D1003), drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds, formed into a female mould (R 89 to 86, a new mould), so its seen face keeps its gloss; it hides the COB and lights evenly




**FRONT**



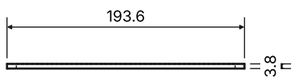

**FROM THE RIGHT**



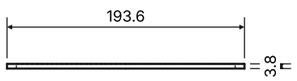

**FROM ABOVE**

**4. rim: lathe, cast-acrylic; 17 cm3 each; 1 for the set; 1:5**

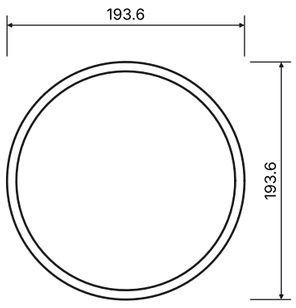
closes the three layers at the mouth, bonded and polished with them; a press-in ePTFE vent in its face keeps the gaps between the layers at room pressure and dry. PROPOSAL 2026-11-03 (the night pictures: a lit polished edge at the mouth pipes the diffuser's light out as a bright blue line): v3's opal ends 4 inside the rim, as the clear shell does, and the rim's exposed face is bead-blasted satin, not polished: no lit polished edge faces the room; about +3*to* + 5 a shade for the blasting [E], the opal's shorter trim free




**FRONT**




**FROM THE RIGHT**




**FROM ABOVE**

## Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

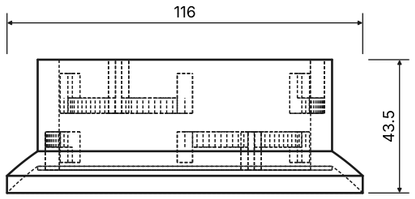
mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

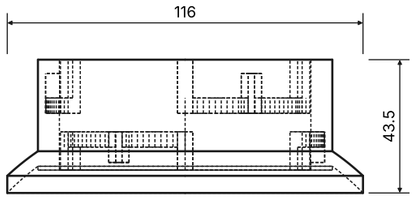
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**5. collar: lathe, aluminium; 82 cm3 each; 1 for the set; 1:2**

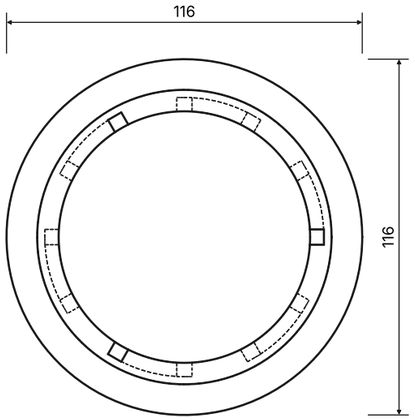
aluminium 6061, black anodised, satin: a tube through the crown with a spherical flange under the skin; three blind L-slots with T-ended notches (the quarter-turn bayonet, 30 deg) work either way up; M96 × 1.5 outside for the clamp ring. It carries the cover mechanically: the bond is a seal only




FRONT



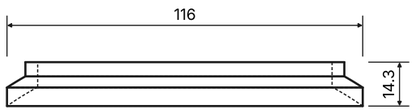

FROM THE RIGHT



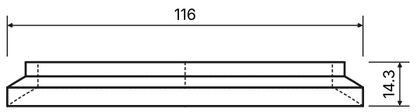

FROM ABOVE

**6. clamp: lathe, aluminium; 24 cm3 each; 1 for the set; 1:2**

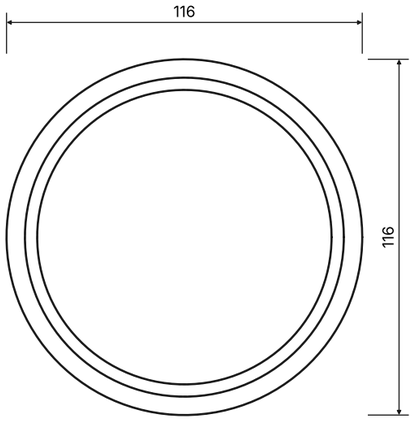
aluminium, black anodised: screwed onto the collar over the crown, its spherical face on a silicone ring, to a torque the shade maker sets (the acrylic is clamped between two silicone rings, never metal on acrylic); no thread-locker near acrylic (review 2026-11-01: anaerobics craze it): held by the silicone's preload and a nylon-tipped set screw through the ring into the collar




FRONT




FROM THE RIGHT




FROM ABOVE

## Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

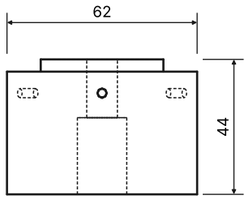
mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

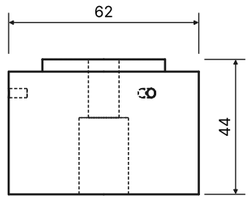
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**9. hub: lathe, aluminium; 119 cm3 each; 1 for the set; 1:2**

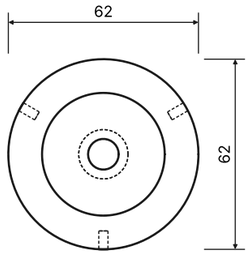
aluminium 6061, black anodised: the core, turned from bar. The COB on its holder at the top of the 46 bore (22 up: the cut-off, about 46 deg from the vertical), the reflector in the bore, the PEEK ring round it, the crown on its neck on a thermal pad; a 10 bore for the wires




FRONT



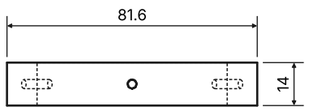

FROM THE RIGHT



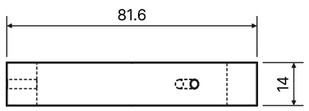

FROM ABOVE

**10. ring: lathe, peek; 31 cm3 each; 1 for the set; 1:2**

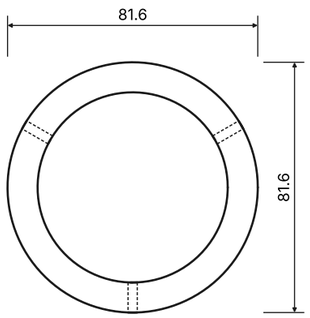
PEEK, unfilled (0.25 W/m K), black: the thermal break. It carries the three pins and the detent, so the cover touches the fixture only through it. Review 2026-11-01: its 81.6 OD is the centering land in the collar's 82 bore (0.2 diametral), three Ø3.2 clearance holes for the pins' threads




FRONT




FROM THE RIGHT




FROM ABOVE

## Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

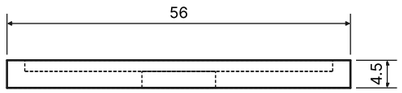
mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

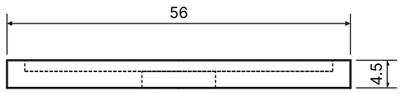
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**13. carrier: lathe, aluminium; 7 cm3 each; 1 for the set; 1:1**

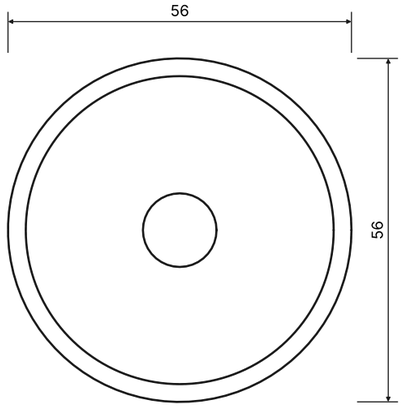
option 3 (2026-11-10): aluminium 6061, black anodised, Ø56 × 4.5, its underside on the rim plane; the board in a Ø50.2 pocket 1.8 deep in its top, so the LEDs' faces sit 4.9 above the rim plane




FRONT



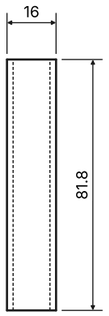

FROM THE RIGHT



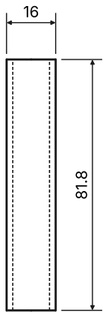

FROM ABOVE

**15. stem: tube, aluminium; 7 cm3 each; 1 for the set; 1:2**

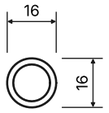
16/12 aluminium tube, black anodised, from the carrier up into the hub's socket, bonded and cross-pinned; the wires up its bore; a heat pipe inside it optional (about 2.5 W of heat)




FRONT




FROM THE RIGHT




FROM ABOVE

## Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

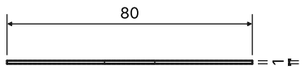
mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

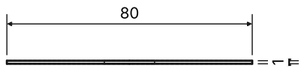
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**16. white-disc: laser, aluminium; 5 cm3 each; 1 for the set; 1:2**

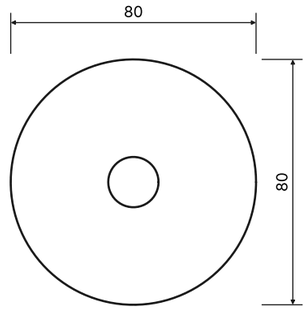
a white Ø80 × 1 disc under the hub (white powder coat), and the collar's flange face inside painted white: the crown's inside white, so the light the shade sends back up returns down




**FRONT**



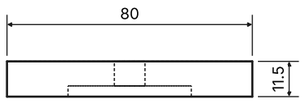

**FROM THE RIGHT**



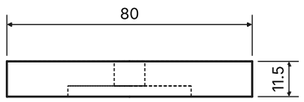

**FROM ABOVE**

**17. crown: lathe, aluminium; 53 cm3 each; 1 for the set; 1:2**

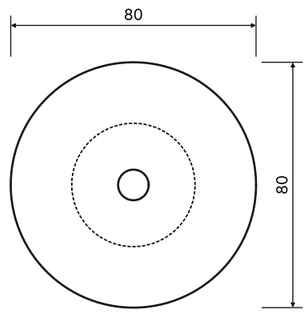
the heat sink (round 41; the fins until then), round 32f's low top: 80 across, 11.5 tall over the collar, turned from 6061 bar, black anodised (emissivity about 0.85: half its loss is radiation), on the core's neck on a thermal pad, 0.5 clear of the PEEK ring and the collar




**FRONT**




**FROM THE RIGHT**




**FROM ABOVE**

## Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

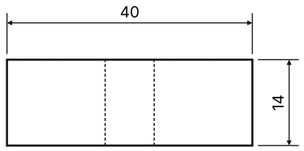
mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

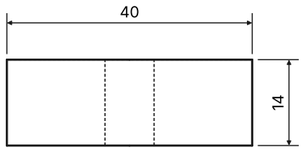
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**18. gland: lathe, aluminium; 17 cm3 each; 1 for the set; 1:1**

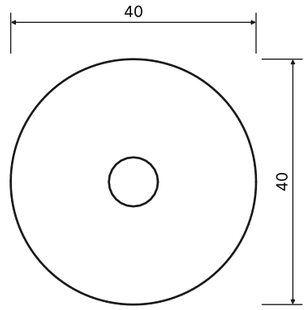
the cap, 40 × 14: aluminium, black anodised and polished (round 32f's was black acrylic: here it runs at the hub's 49 to 52 C), over the listed cord grip (the strain relief, tested with the fixture's weight), on two M3 into the crown; part of the heat sink




FRONT



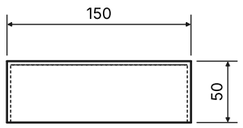

FROM THE RIGHT



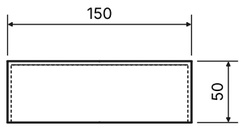

FROM ABOVE

**20. canopy: lathe, cast-acrylic; 118 cm3 each; 1 for the set; 1:5**

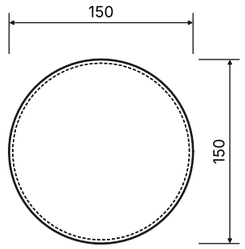
150 × 50, black cast acrylic, polished: a cover only (review 2026-11-01), screwed to the steel strap, which carries the load; the driver and the splices are in the steel enclosure




FRONT




FROM THE RIGHT




FROM ABOVE

## Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

| no. | item | make | qty | what | size | each USD |
|---|---|---|---|---|---|---|
| 1 | shell | formed | 1 | Cast acrylic sheet 1/8 in, TruCast stock tint in the line's code, drape-formed; CUSTOM POURS (2026-11-11): Dust, Acid, Ice the same dye at 3x the stock loading, Smoke at the speakers' own pour (0.3x) | 160 cm3, 210 × 210 × 98 | 1,513.77 |
| 2 | liner | formed | 1 | Cast acrylic sheet, clear (TruCast 0000), drape-formed over an aluminium mould | 265 cm3, 202 × 202 × 89 | 1,519.57 |
| 3 | skin | formed | 1 | Perspex GS 041 3 mm opal, or equal (T 54%, R 39%, ASTM D1003): cast acrylic sheet, volume-diffusing, formed over an aluminium mould | 120 cm3, 188 × 188 × 85 | 1,212.95 |
| 4 | rim | lathe | 1 | Cast acrylic tube, tinted through | 204 × 204 × 14 | 67.88 |
| 5 | collar | lathe | 1 | Aluminium 6061 | 126 × 126 × 53 | 66.22 |
| 6 | clamp | lathe | 1 | Aluminium 6061 | 126 × 126 × 24 | 54.65 |
| 7 | gaskets | buy | 1 | Two moulded silicone rings (Shore 40, black, 1 mm, spherical): one between the collar's flange and the skin inside, one between the clamp ring and the shell outside | | 3.00 |
| 8 | vent | buy | 1 | Gore or alike A press-in ePTFE pressure-equalising vent (about 4 across), in the rim ring: it lets the air gaps between the layers breathe and keeps out dust and liquid | | 1.50 |
| 9 | hub | lathe | 1 | Aluminium 6061 | 72 × 72 × 54 | 52.00 |
| 10 | ring | lathe | 1 | PEEK rod, unfilled | 92 × 92 × 24 | 115.48 |
| 11 | pins | buy | 1 | Three stainless shouldered pins, M3 thread with a Ø4 × 10.5 shoulder, screwed through the PEEK ring into the hub, a pre-applied nylon patch (no anaerobic locker: it crazes acrylic); 4 mm or more of bayonet engagement against a centering | | 6.00 |
| 12 | tether | buy | 1 | Review 2026-11-01: the shade's secondary retention, a 1 mm stainless wire-rope tether (7×7), crimped loops, from an eye screwed into the hub to an eye on the collar, slack in use; rated well over 10 x the 1.1 kg shade | | 4.00 |
| 13 | carrier | lathe | 1 | Aluminium 6061 | 66 × 66 × 14 | 46.58 |
| 14 | board | buy | 1 | Samsung (or Lumileds / Nichia equivalent, to choose) Hemisphere option 3 (the owner, 2026-11-10: "Core cut in half"): a flat Ø50 aluminium-core ring board, 24 mid-power 3030 LEDs (Samsung LM301H EVO class, CRI 90, 2700 K; 3000 K for Ice and | | 14.00 |
| 15 | stem | tube | 1 | Aluminium 6061 | 26 × 26 × 92 | 46.55 |
| 16 | white-disc | laser | 1 | Aluminium 6061 | 90 × 90 × 11 | 47.23 |
| 17 | crown | lathe | 1 | Aluminium 6061 | 90 × 90 × 22 | 49.35 |
| 18 | gland | lathe | 1 | Aluminium 6061 | 50 × 50 × 24 | 46.50 |
| 19 | cord | buy | 1 | any (UL listed) Black textile-braided pendant cord, 2-conductor 18 AWG, UL listed (SVT-rated, or a listed Class 2 cord: past the driver in the canopy it carries the LED's 26 V DC, not mains), 3 m cut to the drop; a listed cord grip in the cap and | | 16.00 |
| 20 | canopy | lathe | 1 | Cast acrylic tube, tinted through | 160 × 160 × 60 | 106.44 |
| 21 | strap | buy | 1 | any (a lighting supplier) Zinc-plated steel crossbar, 1.6 mm, screwed to the 4 in junction box; the cord grip, the suspension gripper and the canopy's screws anchor in it, and the driver and the 120 V splices sit in a closed steel enclosure on it. | | 6.00 |
| 22 | enclosure | buy | 1 | to make (a sheet-metal shop) or a listed driver enclosure Review 2026-11-01: a closed steel enclosure, 92 × 64 × 37, 1.0 wall, zinc-plated, round the driver and the 120 V splices, screwed to the canopy strap; a listed bushing where the Class 2 | | 12.00 |

## Parts list (1 of 2)

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

| no. | item | make | qty | what | size | each USD |
|---|---|---|---|---|---|---|
| 23 | driver | buy | 1 | ERP Power ESS010W-0180-42 (the 1,000 lm level, 2026-11-10; the same driver): constant current 180 mA, 7.6 W, 24-42 V DC out, 120-277 V AC in; forward-phase (TRIAC) and reverse-phase (ELV) dimming at 120 V AC and 0-10 V, 1-100 %; Class 2; | | 16.00 |
| 24 | plunger | buy | 1 | Kipp or Misumi A stainless ball spring plunger M5, light force (about 10 N), threaded into the hub: the bayonet's detent, clicking into a dimple in the collar | | 4.50 |
| 25 | listing | buy | 1 | UL or Intertek (ETL) Listing the pendant to UL 1598 (and the table lamp's battery and charger to UL 153 / IEC 62368): a one-off cost a design, spread over the first run | | [PRICE] |
| 26 | suspension | buy | 1 | [option: suspension] Griplock Systems or Gripple (to choose) Round 43, PROPOSAL (certification.md): a 1.5 mm black-coated stainless suspension cable, 2 m, cut to the drop, with an adjustable gripper in the canopy and a fixed end in the cap; | | 12.00 |

## Parts list (2 of 2)

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

## Notes

1. Option 3 (the owner, 2026-11-10: "it should work like Core cut in half"; the main build): the light sits at the mouth's centre and faces up into the shade, so the shade glows in its tint and the light cast down is the bulb's. A Ø50 ring board of 24 mid-power LEDs (two strings of 12 at about 90 mA, about 1,000 lm, about 200 lux under the lamp) on a black Ø56 × 4.5 carrier, its underside on the rim plane, the LEDs' faces about 5 above it; a 16/12 black anodised stem up to the hub; the crown's inside white (a white Ø80 disc under the hub, the collar's flange face white); the opal to the rim. The COB, its reflector and holder are gone; the ERP driver, the Class 2 cord, the canopy and the UL 1598 path stay. Without a lens until a stock side-emitting ring optic with an IES file is named. Heat test needed: the hub about 6 K warmer than today (estimate). The notes below describe the COB build.
2. Hemisphere (the Dome until round 35; Hemisphere North in rounds 35 to 37): the shade hangs mouth down. Its shades swap by hand: lift the shade 5 mm, turn it 30 degrees and lower it off. To fit one, line up the three slots, push up, turn until the plunger clicks, and let go: its weight holds each pin up in its notch, so a knock cannot turn it off. No tools, and the shade is never live (the COB, the hub and the driver stay up).
3. The shade is clamped, not hung on a glue joint: the collar's flange sits under the crown inside, a ring screws down over it outside, the acrylic between two silicone rings. A failed bond cannot drop it.
4. The wall is tint, opal, tint: a 9 mm tinted shell, a 3 mm volume-diffusing opal liner, a 3 mm tinted skin, each formed from cast sheet on an aluminium mould, nested, closed at the mouth by a tinted rim with a vent. Opal scatters through its depth, so it still diffuses where it touches the tint (a bead-blasted clear liner would go clear wherever it was bonded). Nothing white is seen.
5. Light: a Bridgelux Vero SE 10 (2700 K, CRI 90) at the top of a 46 bore lined with a reflector, 22 up: the cut-off about 46 deg from the vertical, and about 85 % of its light out (the black bore passed 52 %). About 575 lm at the COB (the owner, round 36): about 590 at 180 mA (4.4 W) on an ERP Power ESS010W-0180-42 (Class 2, FCC Part 15 Class B), dimmed 100 % to 1 % by a phase-cut (TRIAC or ELV) dimmer or 0-10 V. Not for 1,100 lm: the low top is sized for 575 (1,100 needs the fins of rounds 35 to 39).
6. Heat (round 41): round 32f's low top is the heat sink, no fins: a black anodised aluminium crown 80 across and 11.5 tall over the collar, and an aluminium cap 40 × 14 over the cord grip, the cap's top 126 above the rim. A PEEK ring carries the bayonet's pins, so the hub reaches the shade only through it and air. At 575 lm the hub makes 2.7 W of heat; estimated in a 35 C room: the hub 49 C, the COB's case 50 C, the collar and the acrylic beside it 41 C (hemisphere_thermal.py; to measure, README round 35).
7. Mains stops at the canopy: the driver is there, and the cord carries the LED's 26 V DC (Class 2). The cord is a listed 18 AWG 2-conductor in a black textile braid, its grips tested with the fixture's weight. To be listed to UL 1598 (hardwired) with its own shades, each tint in the listing; the canopy's strap and steel enclosure carry the grip, the gripper and the splices, the acrylic a cover only.
8. Hang over a bar at 750 to 900 centres, the rim 700 to 800 above the counter; over a table, one a table, the rim about 650 to 750 above it.

## Notes

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-11-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15